

Kaushik Saravanan

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EDUCATION

University of California, Irvine

B.S in Computer Engineering, GPA: 3.87

Irvine, CA

Sep. 2023 – July 2027

PROFESSIONAL EXPERIENCE

Intellinum

AI Research and Development Intern

Dallas, TX

July 2025 – Current

- Developed Python-based MCP servers and tools to integrate with Oracle Fusion ERP, WMS, and OTM applications through REST APIs
- Automated API documentation parsing to populate Qdrant vector database with metadata from 16,000+ endpoints, for semantic searches
- Linked MCP tools to Claude Desktop to create conversational interface for testing

PROJECTS AND EXPERIENCE

Zotbotics | *Project Manager*

September 2024 - Current

- Developed technical curriculum for open-source Arctos robotic arm and spiderbot projects, mentoring 40+ students core in robotics, mechanical assembly, and electrical system fundamentals
- Organized presentations/competitions for members to showcase completed robot arms/spiderbots
- Built a 6-axis BCN3D Moveo robotic arm and 12 servo spider bot using makerspace tools and printers

LLM Controlled Drone

September 2025 - March 2026

- Built quadcopter using Matek F405-WMN flight controller and ArduPilot in 3-person team, flew autonomous 3 waypoint mission in geofence
- Streamed live drone feed from Raspberry Pi Zero W cam to an AWS cloud server running YOLO object-detection, linked via Tailscale VPN
- Wrote MCP server enabling an LLM(Claude) to set flight parameters, flight mode, arm/takeoff, and disarm drone

SAE Aero Design | *Propulsion and Avionics Engineer*

June 2025 – June 2026

- Researched gyroscopic stabilizing systems, and various motor specifications for RC plane
- Performed static thrust tests for multiple propeller configurations using an ESP-32 based thrust stand, analyzing load-cell and current-sensor data to determine optimal prop setup for thrust and flight time

IEEE Open Project Space

October 2024 - June 2025

- Completed a year-long embedded systems course focusing on circuit design, PCB development, and microcontroller programming
- Built 8 unique IoT projects using ESP32s, various sensors/actuators, breadboards, and perfboards with hands-on soldering experience
- Utilized KiCad to design rover remote control PCB

UnlockIt | *LifeTank Competition, 1st Place*

Jan 2024 – March 2024

- Led 5-person team in developing IoT dorm door unlocking prototype
- Organized technical development, delegated roles, and presented to industry expert panel
- Won first place and \$600 prize for innovative hardware solution and market viability

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, SQL, Verilog, HTML/CSS

AI & Software: MCP, REST APIs, OpenAPI, Qdrant Vector Database, Claude API, AWS, Git, Linux, VS Code

Embedded & Robotics: ESP32/Arduino, Raspberry Pi, ArduPilot, Circuit Design, PCB Design (KiCad), Soldering, Sensors, Actuators, Oscilloscopes

Manufacturing & CAD: Onshape, 3D Printing, Laser Cutting, CNC Machining, OrcaSlicer